

STANNARD ROCK, MI, USA

WMO: 994200

Lat: 47.184N

Lon: 87.225W

Elev: 183

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.9	-15.5	-22.4	0.5	-17.0	-20.2	0.6	-14.7	28.1	-3.3	26.4	-3.2	11.6	290	1.023

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	24.2	19.6	22.8	18.9	21.6	18.2	21.4	22.7	20.1	21.9	19.1	21.0	9.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	16.0	22.0	19.5	14.5	20.9	18.3	13.5	20.1	63.1	22.8	58.4	21.8	54.9	21.0	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	18.8	16.6	-19.9	27.3	3.6	2.2	-22.5	28.8	-24.6	30.1	-26.6	31.3	-29.2	32.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-6.3	-6.5	-2.2	2.8	7.9	12.6	16.5	17.4	14.8	8.5	2.3	-2.4
	DBStd	9.29	5.11	5.09	5.15	3.81	3.93	3.63	3.53	2.88	3.48	3.77	4.29	4.11
	HDD10.0	2366	505	461	379	219	91	13	1	0	4	76	231	386
	HDD18.3	4735	763	694	636	467	326	176	78	51	114	306	480	644
	CDD10.0	727	0	0	1	2	24	92	202	231	147	28	1	0
	CDD18.3	55	0	0	0	0	1	4	21	23	7	0	0	0
	CDH23.3	83	0	0	0	0	2	13	31	27	11	0	0	0
	CDH26.7	4	0	0	0	0	0	1	2	1	1	0	0	0
Wind	WSAvg	9.0	10.6	10.5	9.2	8.8	8.8	7.7	7.5	7.3	8.4	9.5	10.2	10.3
Precipitation	PrecAvg	719	54	34	39	55	77	68	62	54	78	75	57	58
	PrecMax	998	139	67	88	115	174	163	168	154	217	208	139	111
	PrecMin	492	13	11	1	16	32	13	23	9	26	20	26	7
	PrecStd	140	32	16	23	28	36	33	30	37	39	42	25	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	4.6	13.8	15.4	22.6	25.3	25.8	25.9	25.0	20.0	12.6	6.6
		MCWB	3.8	3.0	12.5	10.5	17.0	18.1	20.5	20.0	19.7	16.2	10.5	4.5
	2%	DB	3.5	2.4	8.4	12.1	19.1	22.4	24.0	23.9	21.8	17.2	10.2	5.1
		MCWB	2.2	1.0	5.4	8.1	14.3	17.3	19.6	20.5	18.0	13.9	8.2	3.8
	5%	DB	2.3	1.2	5.8	9.5	16.2	20.3	22.7	22.4	20.1	15.1	8.9	4.2
		MCWB	1.3	0.0	3.8	6.1	11.7	15.5	18.9	19.3	17.3	12.8	7.0	3.1
	10%	DB	0.8	0.0	3.8	7.3	14.0	18.6	21.5	21.1	18.9	13.5	7.8	3.2
		MCWB	-0.3	-1.2	2.2	4.4	10.0	14.8	18.3	18.3	16.6	11.7	6.2	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.4	13.1	12.5	19.7	20.7	23.1	23.0	20.5	17.5	10.6	5.3
		MCDB	5.1	4.5	14.9	15.4	21.9	24.2	24.7	23.9	23.1	18.4	12.1	5.5
	2%	WB	2.7	1.5	6.7	8.9	15.7	18.8	21.3	21.6	19.1	14.8	8.6	3.8
		MCDB	3.3	2.3	8.2	11.6	18.1	21.1	22.5	22.5	20.9	16.6	9.5	4.3
	5%	WB	1.5	0.3	4.3	6.8	13.0	17.3	20.1	20.1	18.0	13.2	7.3	3.0
		MCDB	2.4	1.1	5.8	9.4	15.2	20.0	21.7	21.9	19.4	14.6	8.1	3.7
	10%	WB	0.2	-0.9	2.6	5.1	10.5	15.7	19.2	18.9	17.1	11.9	6.3	1.9
		MCDB	1.1	0.1	4.1	7.3	13.2	18.1	21.0	20.8	18.4	13.5	7.3	2.7
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.4	4.6	4.6	6.6	7.8	6.6	4.4	4.1	3.5	3.8	3.8
		MCDBR	5.0	5.2	6.4	8.1	11.1	9.8	7.5	5.8	5.7	5.5	4.6	4.5
		MCWBR	4.8	4.7	4.8	5.1	7.1	6.3	5.0	3.7	3.8	4.1	4.2	4.8
	5% WB	MCDBR	4.7	5.1	6.4	8.3	10.6	9.5	6.7	5.4	5.4	5.4	4.2	4.6
		MCWBR	4.8	5.0	5.3	5.9	8.2	7.5	5.5	4.3	4.3	4.6	4.1	4.7
Clear-Sky Solar Irradiance	taub	0.278	0.291	0.314	0.344	0.379	0.385	0.395	0.375	0.367	0.335	0.305	0.273	
	taud	2.512	2.458	2.413	2.369	2.370	2.376	2.396	2.453	2.458	2.515	2.524	2.543	
	Ebn at Noon	829	887	912	910	886	879	865	870	843	822	784	796	
	Edh at Noon	62	80	98	113	117	117	114	103	94	76	62	54	
All-Sky Solar Radiation	RadAvg	0.92	1.75	3.28	4.61	5.60	6.12	6.18	5.32	3.82	2.22	1.08	0.69	
	RadStd	0.07	0.21	0.40	0.52	0.37	0.45	0.32	0.36	0.23	0.27	0.17	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page